

1. Give asymptotic upper and lower bounds for $T(n)$ in each of the following recurrences. Make your bounds as tight as possible, and justify your answers.

$$T(n) = 2T(n/2) + n^3.$$

$$T(n) = 16T(n/4) + n^2.$$

$$T(n) = 7T(n/3) + n^2.$$

$$T(n) = 7T(n/2) + n^2.$$

$$T(n) = 2T(n/4) + \sqrt{n}.$$

$$T(n) = 3T(n/2) + n \lg n.$$

$$T(n) = 4T(n/2) + n^2 \sqrt{n}.$$

$$T(n) = T(9n/10) + n.$$

2. Give asymptotic upper and lower bounds for $T(n)$ in each of the following recurrences. Make your bounds as tight as possible, and justify your answers. You may assume that $T(1)$ is a constant.

$$T(n) = T(n-1) + n.$$

$$T(n) = T(n-1) + 1/n.$$

$$T(n) = T(n-1) + \lg n.$$

$$T(n) = 2T(n/2) + n \lg n.$$

$$T(n) = 5T(n/5) + n/\lg n.$$